

Experiment 5 Miller Rabin Primality Testing

Code :

```
// C++ program Miller-Rabin primality test
#include <bits/stdc++.h>
using namespace std;
```

```
int power(int x, unsigned int y, int p)
{
    int res = 1;
    x = x % p;
    while (y > 0)
    {
        if (y & 1)
            res = (res*x) % p;
        y = y>>1;
        x = (x*x) % p;
    }
    return res;
}
```

```
bool miillerTest(int d, int n)
{
    int a = 2 + rand() % (n - 4);
    int x = power(a, d, n);

    if (x == 1 || x == n-1)
        return true;

    while (d != n-1)
    {
```

```

        x = (x * x) % n;
        d *= 2;

        if (x == 1) return false;
        if (x == n-1) return true;
    }

    return false;
}

bool isPrime(int n, int k)
{
    if (n <= 1 || n == 4) return false;
    if (n <= 3) return true;

    int d = n - 1;
    while (d % 2 == 0)
        d /= 2;

    for (int i = 0; i < k; i++)
        if (!miillerTest(d, n))
            return false;

    return true;
}

int main()
{
    int k = 4,x;
    cout<<"Enter an INPUT ---->" <<endl;
    cin>>x ;

```

```

        cout << "All primes smaller than "<<x<<" are : \n";
        for (int n = 1; n < x; n++)
            if (isPrime(n, k))
                cout << n << " ";

        return 0;
}

```

OUTPUT :



